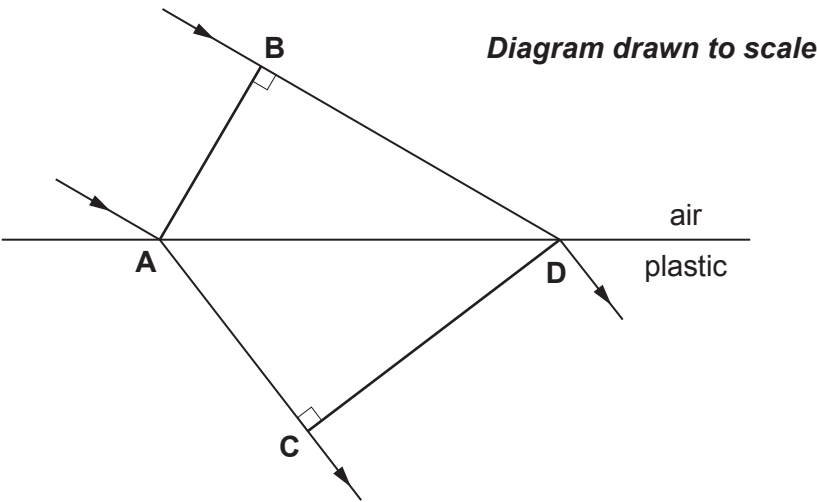


7. (a) The diagram (**real size**) shows a beam of light passing from air into a clear plastic. AB is a wavefront about to enter the plastic, and CD is its position a short time later.



- (i) Determine the speed of light in the plastic **using measurements made with a ruler** from the diagram, and taking the speed of light in air to be equal to its speed in a vacuum. Show your working clearly. [3]

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- (ii) Calculate the refractive index of the plastic. [1]

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(b) In a multimode fibre of length 120.00 m, the length of the longest possible (zig-zag) route for light to travel successfully is 120.90 m. The refractive index of the fibre core is 1.520. Very short pulses of light are sent into one end of the fibre at intervals of 4.0 ns.

(i) Evaluate whether or not overlap (or overtaking) of pulses will occur before the pulses have reached the other end of the fibre. [3]

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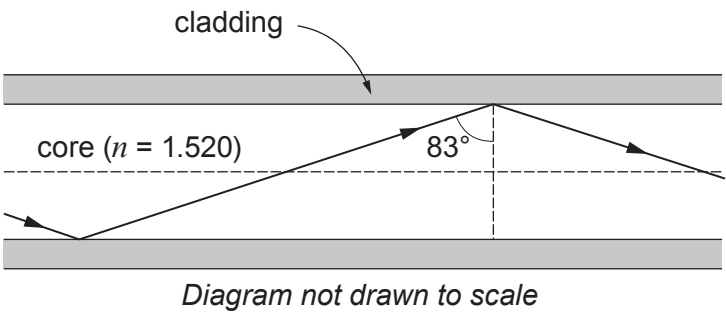
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(ii) Part of the longest possible successful zig-zag route is shown below.



Calculate the refractive index of the cladding. [2]

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END OF PAPER

